QUESTION 1: 4-Bit Fuel Sensor Combinational Logic Circuit [15 Marks]

Problem Summary: A vehicle fuel sensor outputs a 4-bit binary level $G = (G_3G_2G_1G_0)_2$ from 0000_2 (0_{10}) to 1100_2 (12_{10}). Design the Combinational Logic (CL) circuit driving outputs F_1, F_2, F_4, F_8, and R using **five** 2×4 **decoders with enable** (EN):

- **Above 11** ($G = 12$)**:** $F_1, F_2, F_4, F_8 = 1, R = 0$
- **11 to** $> 75\%$ ($G = 10, 11$)**:** $F_1, F_2, F_4 = 1$
- **75% to** $> 50\%$ ($G = 7, 8, 9$)**:** $F_1, F_2 = 1$
- **50% to** $> 25\%$ ($G = 4, 5, 6$)**:** $F_1 = 1$ only
- $\leq 25\%$ ($G = 2, 3$)**:** All outputs $= 0$
- **Below 2** ($G = 0, 1$)**:** $R = 1$ only, all $F_i = 0$.

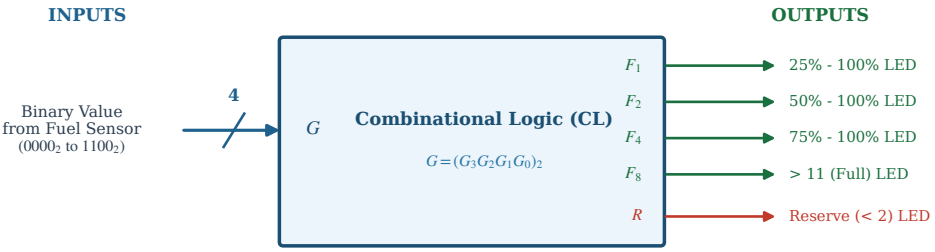


Figure 1.1: Top-level block diagram of the 4-bit Fuel Sensor Combinational Logic (CL) circuit.

Step 1: Fuel Percentage Thresholds & Complete Truth Table

Full tank is $G_{\max} = 12_{10}$ (1100_2), so the percentage thresholds are: **75**% of $12 = \mathbf{9}$, **50**% of $12 = \mathbf{6}$, **25**% of $12 = \mathbf{3}$. Sensor codes $13, 14, 15$ $(1101_2$–$1111_2)$ never occur and are treated as **Don't-Care** (X).

Minterm	G_3	G_2	G_1	G_0	% Level	Fuel Tank Condition	F_1	F_2	F_4	F_8	R
m_0 (0)	0	0	0	0	0.0%	Below 2 ($G < 2$): **Reserve ON** ($R = 1$)	0	0	0	0	**1**
m_1 (1)	0	0	0	1	8.3%	Below 2 ($G < 2$): **Reserve ON** ($R = 1$)	0	0	0	0	**1**
m_2 (2)	0	0	1	0	16.7%	$\leq 25\%$ ($2 \leq G \leq 3$): All Cleared (0)	0	0	0	0	0
m_3 (3)	0	0	1	1	25.0%	$\leq 25\%$ ($2 \leq G \leq 3$): All Cleared (0)	0	0	0	0	0
m_4 (4)	0	1	0	0	33.3%	$> 25\%$ to 50% ($4 \leq G \leq 6$): F_1 **ON**	**1**	0	0	0	0
m_5 (5)	0	1	0	1	41.7%	$> 25\%$ to 50% ($4 \leq G \leq 6$): F_1 **ON**	**1**	0	0	0	0
m_6 (6)	0	1	1	0	50.0%	$> 25\%$ to 50% ($4 \leq G \leq 6$): F_1 **ON**	**1**	0	0	0	0
m_7 (7)	0	1	1	1	58.3%	$> 50\%$ to 75% ($7 \leq G \leq 9$): F_1, F_2 **ON**	**1**	**1**	0	0	0
m_8 (8)	1	0	0	0	66.7%	$> 50\%$ to 75% ($7 \leq G \leq 9$): F_1, F_2 **ON**	**1**	**1**	0	0	0
m_9 (9)	1	0	0	1	75.0%	$> 50\%$ to 75% ($7 \leq G \leq 9$): F_1, F_2 **ON**	**1**	**1**	0	0	0
m_{10} (10)	1	0	1	0	83.3%	$> 75\%$ to 11 ($10 \leq G \leq 11$): F_1, F_2, F_4 **ON**	**1**	**1**	**1**	0	0
m_{11} (11)	1	0	1	1	91.7%	$> 75\%$ to 11 ($10 \leq G \leq 11$): F_1, F_2, F_4 **ON**	**1**	**1**	**1**	0	0
m_{12} (12)	1	1	0	0	100.0%	Above 11 ($G = 12$): F_1, F_2, F_4, F_8 **ON**	**1**	**1**	**1**	**1**	0
m_{13} (13)	1	1	0	1	—	Unused Input (Don't-Care X)	X	X	X	X	X
m_{14} (14)	1	1	1	0	—	Unused Input (Don't-Care X)	X	X	X	X	X
m_{15} (15)	1	1	1	1	—	Unused Input (Don't-Care X)	X	X	X	X	X

Step 2: Canonical Minterm Equations (for Decoder Synthesis) & K-Maps

From the Truth Table, each output is the OR-sum of its active minterms (cascaded cleanly to minimize OR gate fan-in):

$$\mathbf{R = m_0 + m_1}, \quad \mathbf{F_8 = m_{12}}, \quad \mathbf{F_4 = m_{10} + m_{11} + F_8}, \quad \mathbf{F_2 = m_7 + m_8 + m_9 + F_4}, \quad \mathbf{F_1 = m_4 + m_5 + m_6 + F_2}$$

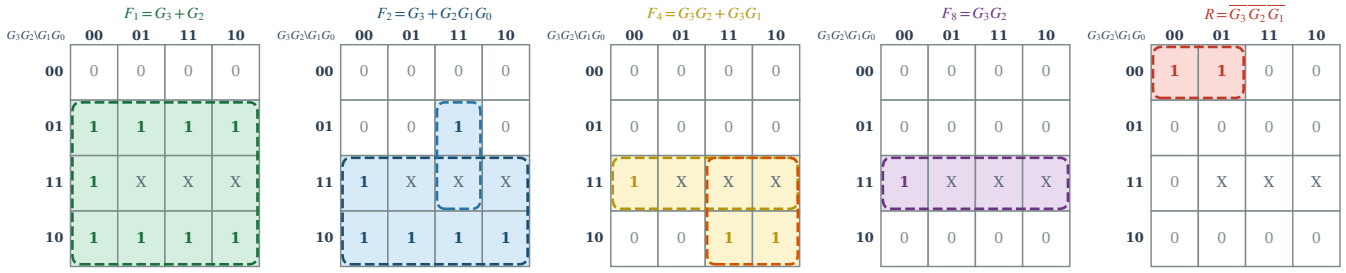


Figure 1.2: 4-Variable Karnaugh Maps giving simplified gate-level SOP equations for F_1, F_2, F_4, F_8, and R.

Step 3: Logic Diagram Using Five 2×4 Decoders with Enable (EN)

- **Decoder D_0 (Stage 1 Bank Selector):** Receives MSBs $A_1 = G_3$ and $A_0 = G_2$ with $EN = 1$. Its outputs Y_0, Y_1, Y_2, Y_3 enable exactly one Stage-2 decoder at a time: $Y_0 \to EN(D_1)$ for $m_0..m_3$; $Y_1 \to EN(D_2)$ for $m_4..m_7$; $Y_2 \to EN(D_3)$ for $m_8..m_{11}$; $Y_3 \to EN(D_4)$ for $m_{12}..m_{15}$.
- **Decoders D_1, D_2, D_3, D_4 (Stage 2 Minterm Generators):** All four share LSBs $A_1 = G_1$ and $A_0 = G_0$ to generate $m_0..m_{15}$.
- **Output OR Gates:** Combine active minterms in a clean forward staircase with **zero wire crossings** on the output side: $\text{OR}_R = m_0 + m_1$, $F_8 = m_{12}$, $\text{OR}_{F4} = m_{10} + m_{11} + F_8$, $\text{OR}_{F2} = m_7 + m_8 + m_9 + F_4$, $\text{OR}_{F1} = m_4 + m_5 + m_6 + F_2$.

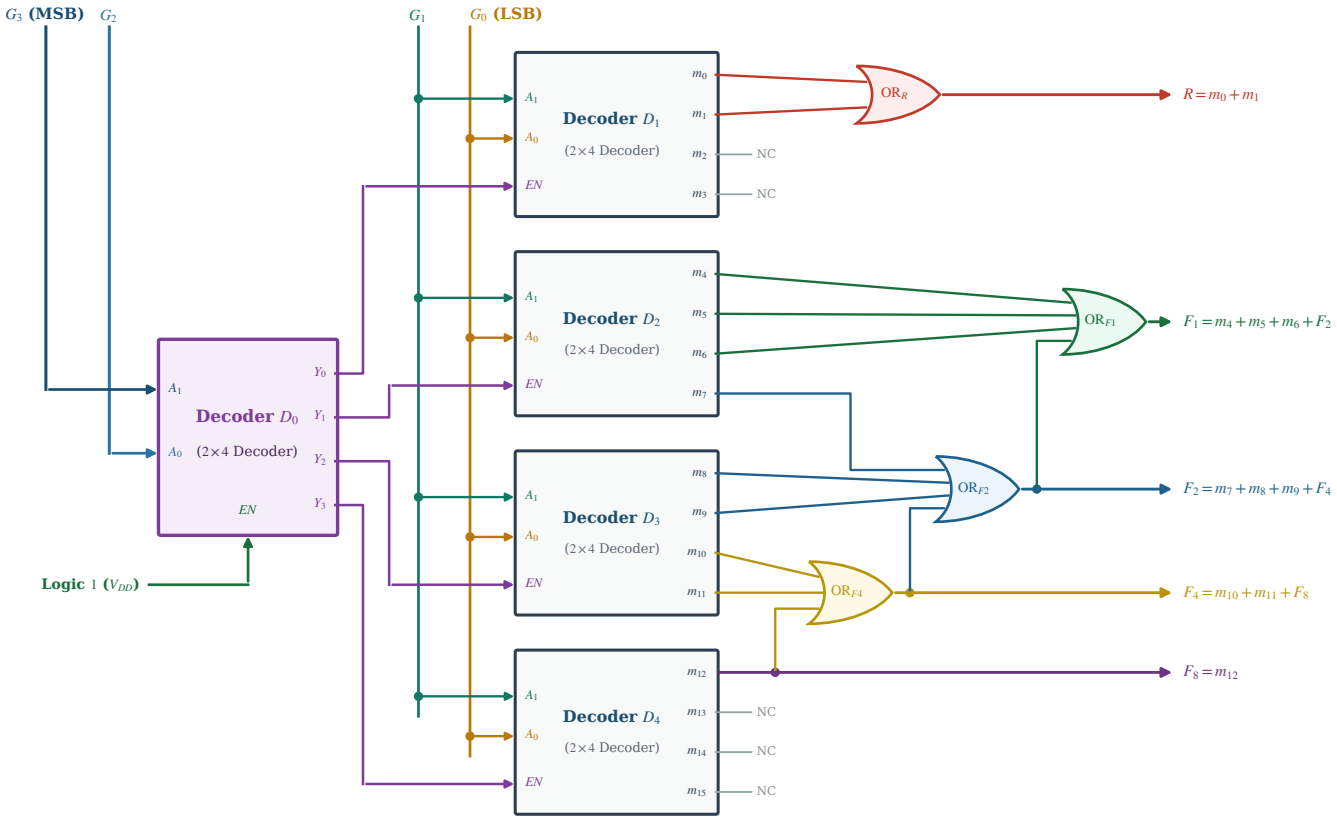


Figure 1.3: Complete Logic Diagram of the Fuel Sensor Combinational Circuit Using Five 2×4 Decoders with Enable (D_0–D_4).

Quick Check Table of Decoder Operation:

Input Range G	Active D_0 Line	Enabled Stage-2	Active Minterm Line(s)	Active Output(s)
$0000_2, 0001_2$ (0, 1)	$Y_0 = 1$	Decoder D_1	m_0 or m_1	$R = 1$ (All $F_i = 0$)
$0010_2, 0011_2$ (2, 3)	$Y_0 = 1$	Decoder D_1	m_2 or m_3 (NC)	**All Outputs** $= 0$
0100_2–0110_2 (4, 5, 6)	$Y_1 = 1$	Decoder D_2	m_4, m_5, or m_6	$F_1 = 1$
0111_2–1001_2 (7, 8, 9)	Y_1 or $Y_2 = 1$	D_2 or D_3	m_7, m_8, or m_9	$F_1 = 1, F_2 = 1$
$1010_2, 1011_2$ (10, 11)	$Y_2 = 1$	Decoder D_3	m_{10} or m_{11}	$F_1 = 1, F_2 = 1, F_4 = 1$
1100_2 (12)	$Y_3 = 1$	Decoder D_4	m_{12}	$F_1, F_2, F_4, F_8 = 1$

QUESTION 2: Automatic Door Lock Digital System & FSM Controller [15 Marks]

Problem Summary: Design an automatic door lock controller with active-high inputs **FS** (Front gate switch), **MS** (Motion sensor), **R** (Async reset), **DC** (Door close switch) and active-high outputs **FM** (Front melody), **MM** (Motion melody), **DO** (Door open):

- **Reset** ($R=1$)**:** Asynchronously goes to S_{init} from any state $\Longrightarrow$ clears $FM=0, MM=0, DO=0$, then moves unconditionally to S_{wait}.
- **In** S_{wait} (FM=0, MM=0, DO=0)**:** If $MS=1$ (priority over FS), go to S_{motion}. Else if $\overline{MS}\cdot FS=1$, go to S_{front}. Else stay in S_{wait}.
- **In** S_{front} (FM=1, MM=0, DO=1)**:** Stay in S_{front} while $DC=0$; return to S_{wait} when $DC=1$ (DC has no effect in other states).
- **In** S_{motion} (FM=0, MM=1, DO=1)**:** Keep $MM=1, DO=1$ ON for **two more clock periods**, then return to S_{wait}.

Step 1: System Block Diagram



Figure 2.1: Top-level Block Diagram of the Automatic Door Lock Controller showing all inputs and outputs.

Step 2: Capturing the Finite State Machine (Both Standard Representations)

Depending on whether the 2 extra clock periods in S_{motion} are tracked by a 2-bit timer T or by unrolling S_{motion} into 3 consecutive clock-cycle states ($S_{\text{motion1}}, S_{\text{motion2}}, S_{\text{motion3}}$), both equivalent Moore FSM diagrams are shown in Figure 2.2:

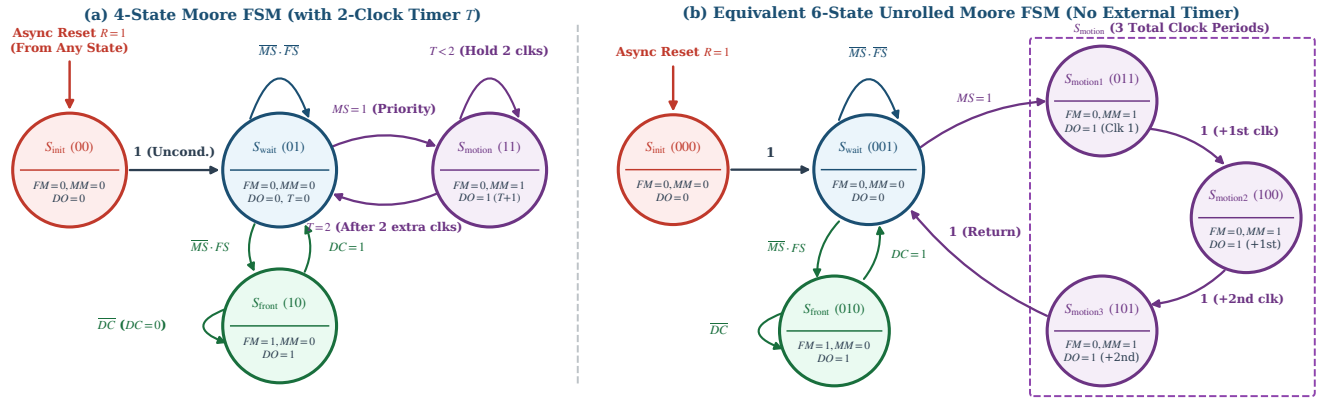


Figure 2.2: Moore FSM State Diagrams: (a) 4-State FSM with 2-Clock-Period Timer T; (b) 6-State Unrolled Pure Moore FSM.

Step 3A: State Table for 4-State FSM with Timer T (Figure 2.2a)

In the 4-state model $(S_1S_0)_2$, state codes are $S_{\text{init}}=00, S_{\text{wait}}=01, S_{\text{front}}=10, S_{\text{motion}}=11$:

Present State	S_1	S_0	Input / Timer Cond.	Next State	S_1^+	S_0^+	FM	MM	DO
S_{init}	0	0	Unconditional (1)	S_{wait}	0	1	0	0	0
S_{wait}	0	1	$\overline{MS}\cdot\overline{FS}$	S_{wait} ($T\leftarrow 0$)	0	1	0	0	0
S_{wait}	0	1	$\overline{MS}\cdot FS=1$	S_{front}	1	0	0	0	0
S_{wait}	0	1	$MS=1$ (Priority)	S_{motion} ($T\leftarrow 0$)	1	1	0	0	0
S_{front}	1	0	$DC=0$ ($\overline{DC}$)	S_{front}	1	0	1	0	1
S_{front}	1	0	$DC=1$	S_{wait}	0	1	1	0	1
S_{motion}	1	1	$T<2$ ($T=0,1$)	S_{motion} ($T\leftarrow T+1$)	1	1	0	1	1
S_{motion}	1	1	$T=2$ (2 extra clks done)	S_{wait} ($T\leftarrow 0$)	0	1	0	1	1

- **4-State Equations:** $S_1^+ = \overline{S_1}S_0(MS+FS)+S_1\overline{S_0}\,\overline{DC}+S_1S_0\overline{(T=2)}$, $\quad S_0^+ = \overline{S_1}\,\overline{S_0}+\overline{S_1}S_0(MS+\overline{FS})+S_1\overline{S_0}DC+S_1S_0$, $\quad FM = S_1\overline{S_0}$, $\quad MM = S_1S_0$, $\quad DO = S_1$.

Step 3B: State Transition Table & Boolean Equations for 6-State Pure FSM (Figure 2.2b)

When implemented as a pure FSM without an external counter, the 6 states are encoded with 3 flip-flops $(Q_2Q_1Q_0)_2$, with unused states $110_2, 111_2$ treated as Don't-Care (X). Assigning $S_{\text{init}} = 000_2$ allows asynchronous reset $R = 1$ to directly clear all flip-flops.

Present State	Q_2	Q_1	Q_0	MS	FS	DC	Next State	D_2	D_1	D_0	FM	MM	DO
S_{init}	0	0	0	X	X	X	S_{wait}	0	0	1	0	0	0
S_{wait}	0	0	1	0	0	X	S_{wait}	0	0	1	0	0	0
S_{wait}	0	0	1	0	1	X	S_{front}	0	1	0	0	0	0
S_{wait}	0	0	1	1	X	X	S_{motion1}	0	1	1	0	0	0
S_{front}	0	1	0	X	X	0	S_{front}	0	1	0	**1**	0	**1**
S_{front}	0	1	0	X	X	1	S_{wait}	0	0	1	**1**	0	**1**
S_{motion1} **(Clk 1)**	0	1	1	X	X	X	S_{motion2}	1	0	0	0	**1**	**1**
S_{motion2} **(+1 Clk)**	1	0	0	X	X	X	S_{motion3}	1	0	1	0	**1**	**1**
S_{motion3} **(+2 Clk)**	1	0	1	X	X	X	S_{wait}	0	0	1	0	**1**	**1**
Unused (110, 111)	1	1	X	X	X	X	**Don't-Care** (X)	X	X	X	X	X	X

Minimized Next-State Equations (D_2, D_1, D_0) and Output Equations (FM, MM, DO):

$$\mathbf{D_2} = \overline{Q_2}\,Q_1\,Q_0 + Q_2\,\overline{Q_1}\,\overline{Q_0} = \mathbf{Q_1\,Q_0} + \mathbf{Q_2\,\overline{Q_0}}$$

$$\mathbf{D_1} = \overline{Q_2}\,\overline{Q_1}\,Q_0\,(MS + \overline{MS}\cdot FS) + \overline{Q_2}\,Q_1\,\overline{Q_0}\,\overline{DC} = \mathbf{\overline{Q_2}\,\overline{Q_1}\,Q_0\,(MS + FS)} + \mathbf{Q_1\,\overline{Q_0}\,\overline{DC}}$$

$$\mathbf{D_0} = \overline{Q_2}\,\overline{Q_1}\,\overline{Q_0} + \overline{Q_2}\,\overline{Q_1}\,Q_0\,(MS + \overline{FS}) + \overline{Q_2}\,Q_1\,\overline{Q_0}\,DC + Q_2\,\overline{Q_1} = \mathbf{Q_2 + \overline{Q_1}\,\overline{Q_0} + \overline{Q_1}\,Q_0\,(MS + \overline{FS})} + \mathbf{Q_1\,\overline{Q_0}\,DC}$$

$$\mathbf{FM} = \overline{Q_2}\,Q_1\,\overline{Q_0} = \mathbf{Q_1\,\overline{Q_0}}, \qquad \mathbf{MM} = \overline{Q_2}\,Q_1\,Q_0 + Q_2\,\overline{Q_1} = \mathbf{Q_2 + Q_1\,Q_0}, \qquad \mathbf{DO} = \mathbf{Q_2 + Q_1} = \mathbf{FM + MM}$$

Step 4: FSM Controller Hardware Architecture Diagram

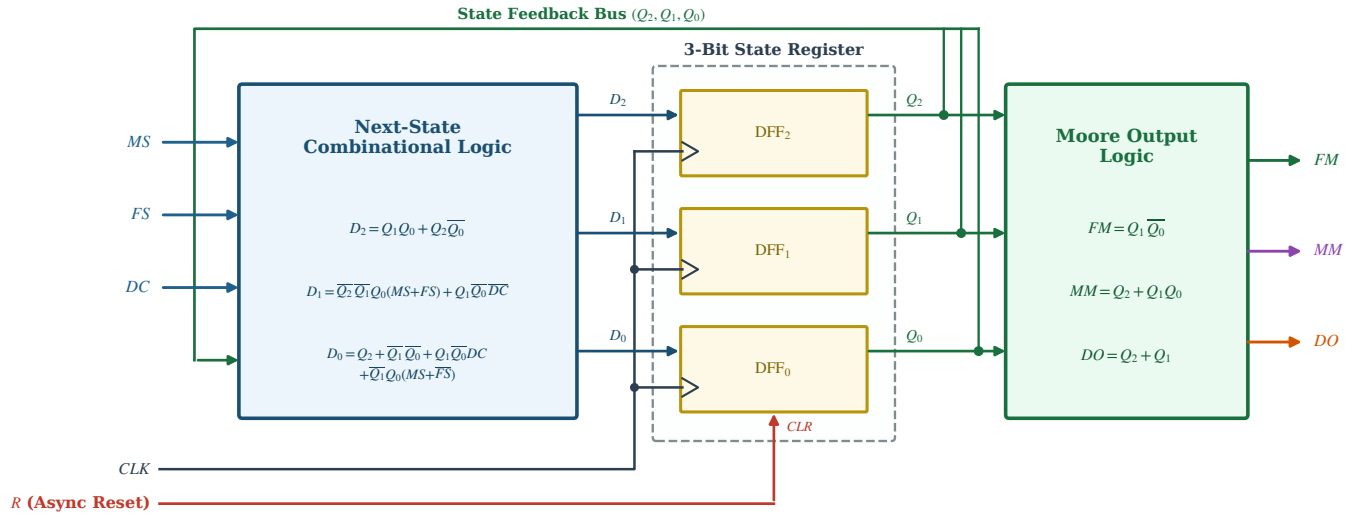


Figure 2.3: Complete Hardware Architecture of the Automatic Door Lock FSM Controller (Next-State Logic, 3-Bit State Register with Async Reset R, and Moore Output Logic).